DEEP LEARNING: ADVANCED MODELS AND METHODS

Knowledge Distillation for Mobile Action Recognition

Compressing a 3D ResNet-50 into an ultra-lightweight MobileNet3D on UCF-101 via logit-based, attention transfer, and cross-frame distillation.

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TRACK
Track 6 - KD for Mobile Action Recognition

INSTITUTION
Università degli Studi di Catania - DMI

PROFESSOR
Prof. Antonino Furnari

REPOSITORY
github.com/danii909/ResNet-to-MobileNet-Action

PROBLEM & MOTIVATION

The Capacity Gap

Goal: Compress a 3D ResNet-50 teacher (pretrained on Kinetics-400) into a MobileNet3D student for on-device video action recognition on UCF-101 (101 classes).

TEACHER

3D ResNet-50

88.82%

TEST ACCURACY

~128 MB | 6.21 ms GPU | 76.6 ms CPU

ACCURACY GAP

−29.3 pp

→

SIZE REDUCTION

13.5×

→

INFERENCE SPEED

1.82×

1.66× CPU (BS=1)

→

STUDENT

MobileNet3D

59.48%

TEST ACCURACY

~9.5 MB | 3.42 ms GPU | 46.1 ms CPU

EXPERIMENTAL SETUP

UCF-101 & Group-Aware Split

101	13,320	24f
ACTION CLASSES	VIDEO CLIPS	@ 112×112 PX

Original Split Structure

UCF-101 is officially divided only into Train and Test partitions, **lacking an actual validation set** required for proper model selection.

⚠️ Data Leakage Risk

Correlated clips from the same source video share actor & background. Random splits cause **spatial leakage** and inflated accuracy.

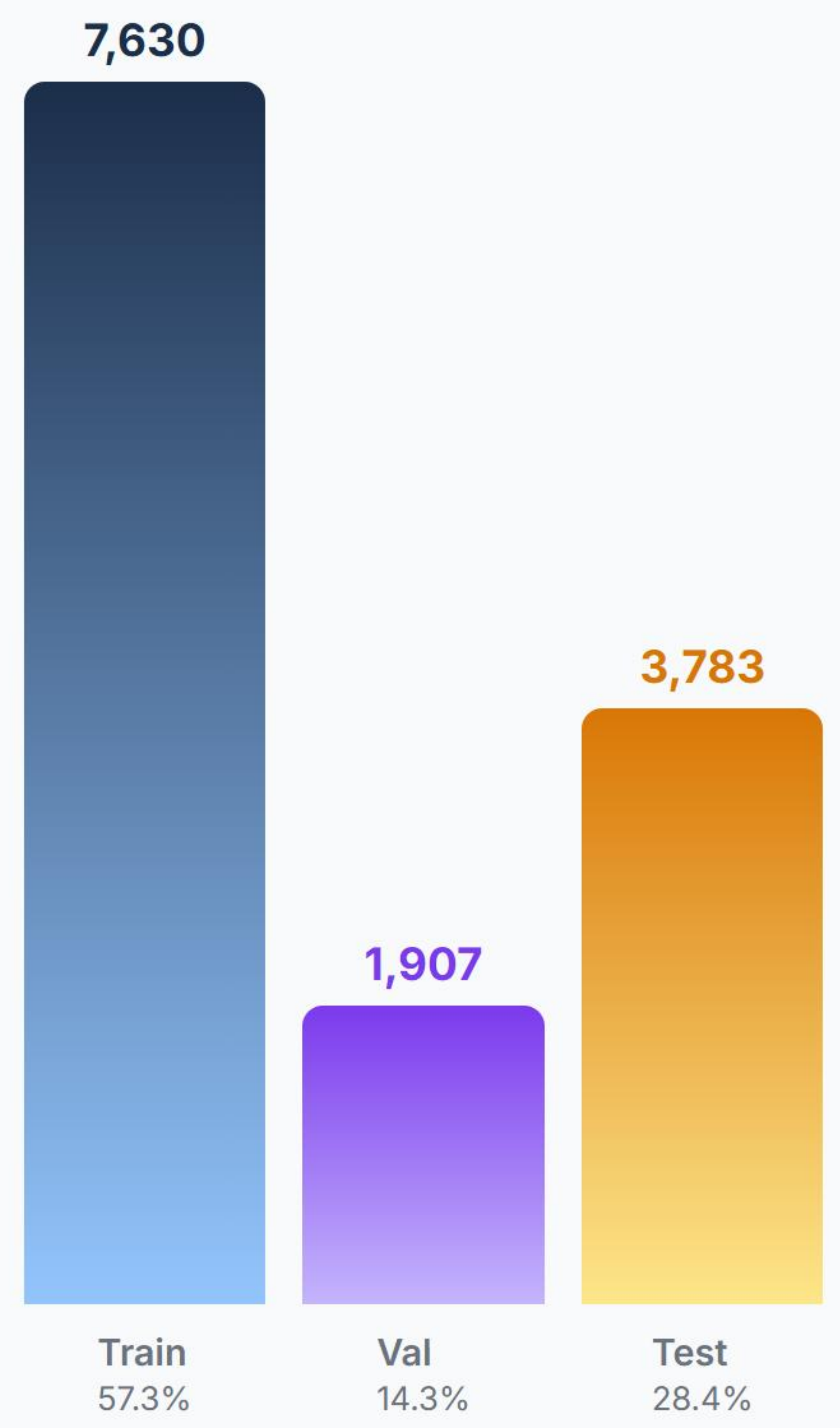
✅ Group-Aware Split

Isolation by group ID (**_gXX_**). No source video shared across Train / Val / Test.

⏱️ Temporal Sampling & Stride

Segmented sampling divides the video into N segments, randomly sampling one frame per segment in training (uniform for eval). Stride is dynamic ($\approx L / N$ frames) to span the entire clip.

NEW SPLIT DISTRIBUTION



THEORY & FORMULATION

Knowledge Distillation Loss

Balanced objective function for knowledge transfer

$$\mathcal{L}_{\text{total}} = \underbrace{\alpha \cdot T^2 \cdot D_{\text{KL}}\left(\sigma(z^S / T) \parallel \sigma(z^T / T)\right)}_{\mathcal{L}_{\text{soft}}} + \underbrace{(1 - \alpha) \cdot \mathcal{L}_{\text{CE}}(z^S, y)}_{\mathcal{L}_{\text{hard}}}$$

$\alpha = 0.7$

TARGET BALANCING

Always set to **0.7** for the majority of experiments to prioritize mimicking the Teacher's soft targets (70%) over the hard ground truth labels (30%).

$T \in \{1, 5, 8, 10, 20\}$

LOGIT SCALING

The main hyperparameter varied across experiments to control the smoothness of probability distributions, extracting latent inter-class similarities.

FAILURE CASES

Augmentation Trade-off & Data Leakage

All tests: T=10, a=0.7, 24f · Why light augmentation was chosen

1. Minimal Augmentation

WITH LEAK	NO LEAK
65.7%	59.3%
Top-5: 87.8%	Top-5: 83.6%

CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Disabled (0.0)
- **Random Erasing:** Disabled
- **Temporal stride:** Fixed (max_stride=1)

2. Light Augmentation ✓

WITH LEAK	NO LEAK
69.5%	64.5%
Top-5: 90.3%	Top-5: 87.4%

CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Light (0.1)
- **Random Erasing:** 5% probability
- **Temporal stride:** Fixed (max_stride=1)

3. Strong Augmentation

WITH LEAK	NO LEAK
55.83%	50.4%
Top-5: 83.27%	Top-5: 78.5%

CONFIGURATION

- **Spatial crop:** Resized Crop (0.6—1.0)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Strong (0.2)
- **Random Erasing:** 15% probability
- **Temporal stride:** Variable (max_stride=2)

QUANTITATIVE RESULTS

Temperature Sweep & Main Result

Systematic analysis of Student behavior as T varies · a = 0.7 fixed

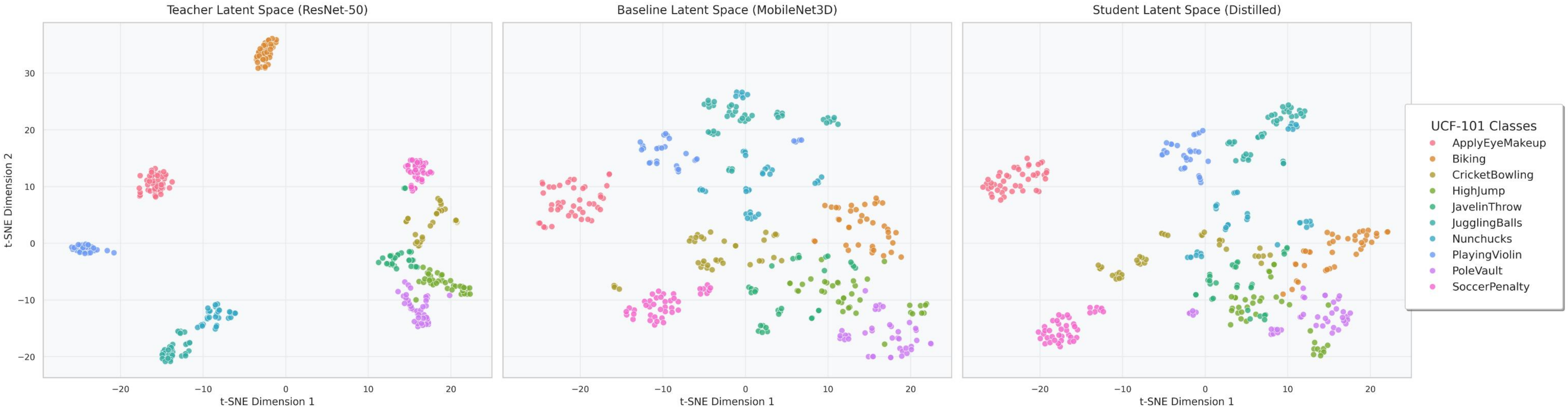
TEMPERATURE	BEST VAL ACC	EPOCH	TRAIN ACC	TEST TOP-1	TEST TOP-5	Δ VS BASELINE
Minimal Augmentation (no KD)	59.18%	—	99.81%	59.48%	82.77%	ref.
T = 1	62.75%	60	98.44%	62.91%	85.59%	+3.43%
T = 5	63.12%	57	95.87%	62.07%	87.02%	+2.59%
T = 8	64.57%	56	97.59%	64.31%	87.68%	+4.83%
T = 10	64.85%	60	98.28%	64.47%	87.44%	+4.99%
T = 20	**65.18%**	**59**	**99.05%**	**65.85%**	**87.81%**	**+6.37%**

TEST ACCURACY VS. TEMPERATURE TREND

QUALITATIVE ANALYSIS

Latent Space Analysis via t-SNE

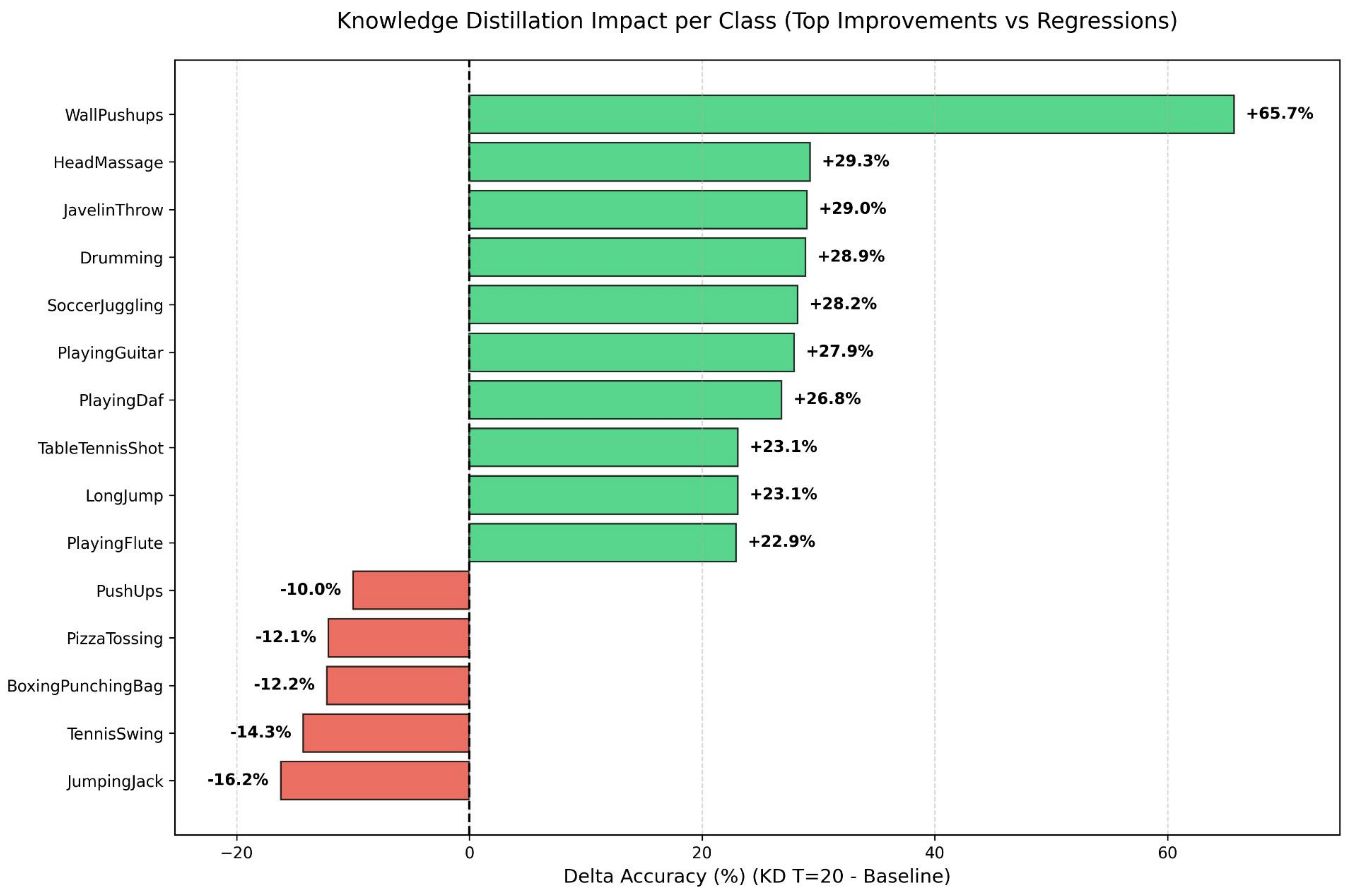
2D projections of pre-classifier embeddings — 10 selected classes



The distilled model (T=20) shows significantly tighter and more distinct class clusters compared to the baseline student, successfully inheriting the well-separated semantic boundaries of the teacher model.

QUALITATIVE ANALYSIS

Per-Class Accuracy: KD T=20 vs Baseline



Improvements

Spatially complex classes (e.g. **WallPushups**, **HeadMassage**) gain most from the Teacher's visual filters.

Regressions

Fast, cyclic actions (**JumpingJack**, **TennisSwing**) suffer from temporal over-smoothing at high T.

Shared Challenges

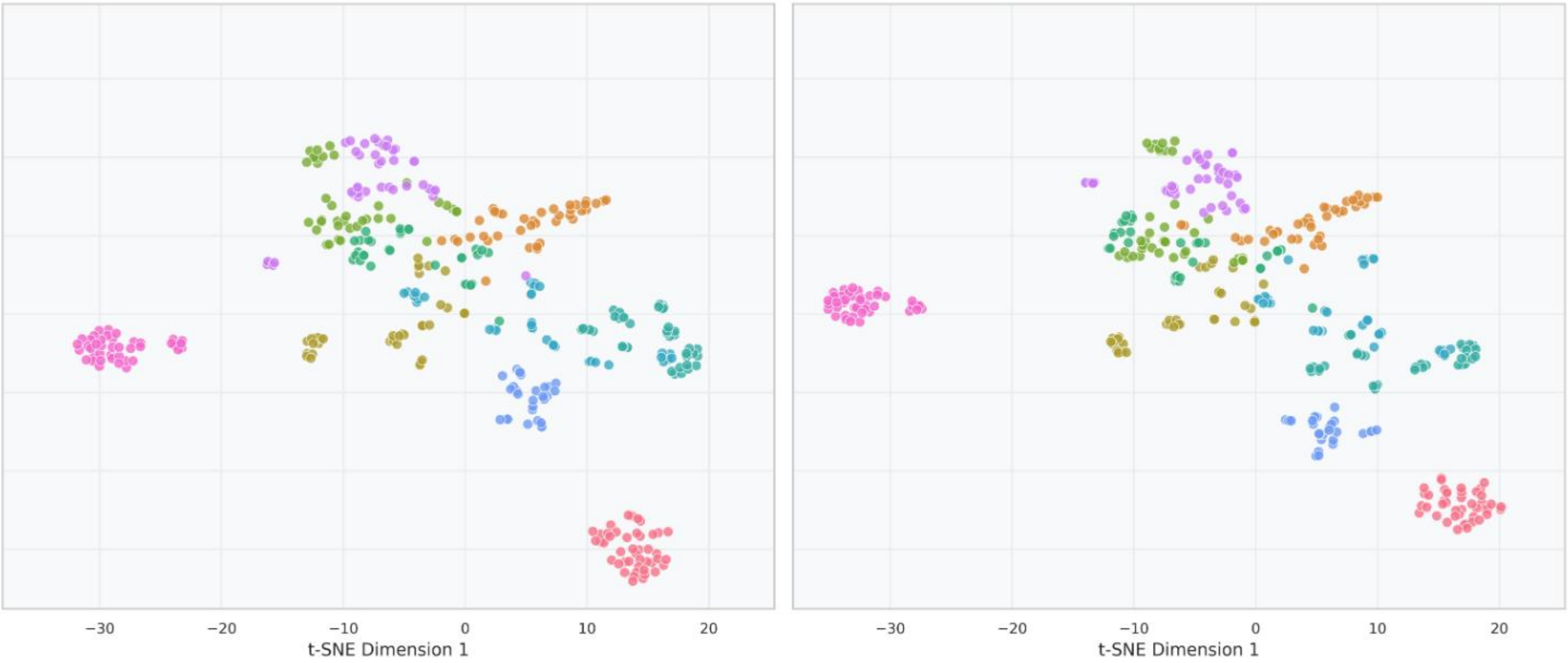
Classes with high structural ambiguity (e.g. **HandstandWalking**, **Nunchucks**) or strong mutual confusion (e.g. **MoppingFloor**) remain unresolved in both models.

ADVANCED METHODS

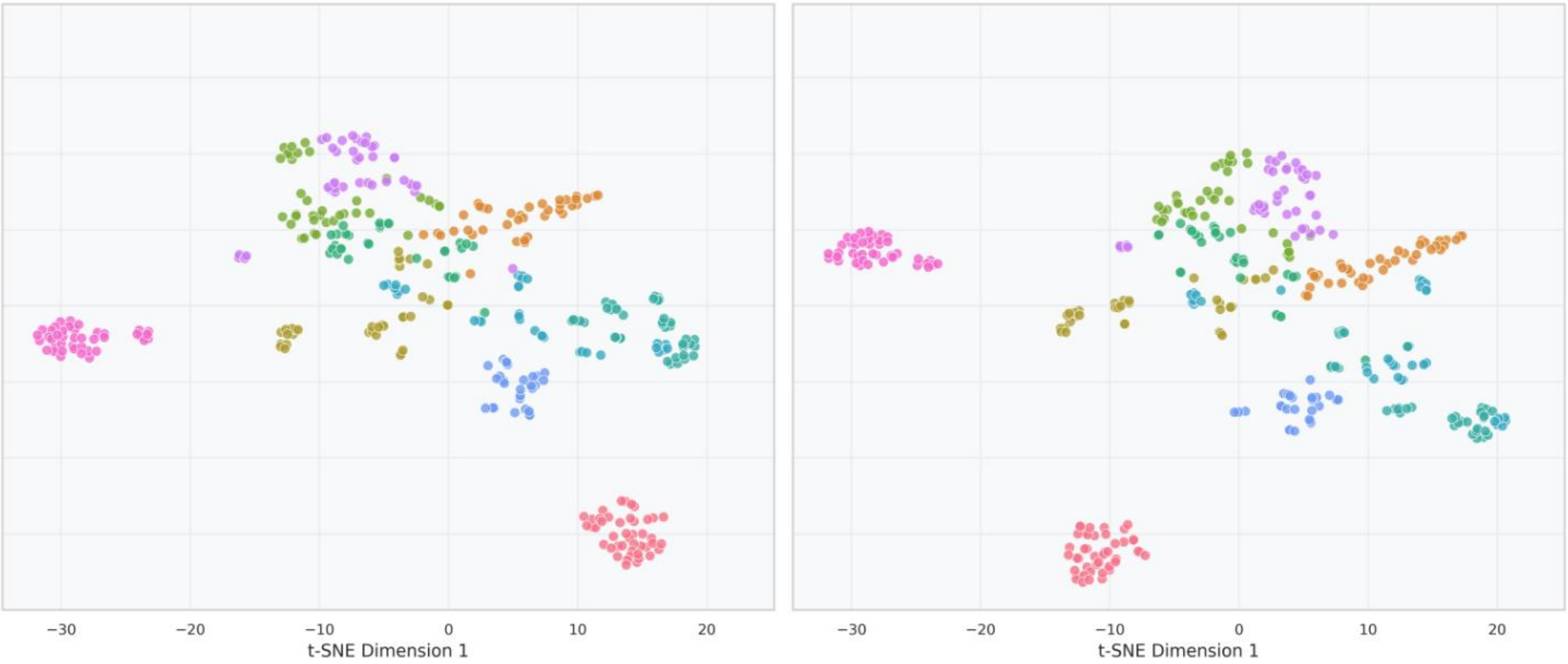
Attention Transfer (AT)

Capacity gap impact on intermediate feature alignment

PURE KD T=20 | **SYMMETRIC AT**



PURE KD T=20 | **TEMPORAL-ONLY AT**



PURE KD T=20
65.85%

CAPACITY LOSS
−1.27%

SYMMETRIC AT
64.58%

Forced spatial alignment acts as noise due to Student's depthwise convolution capacity limit.

PURE KD T=20
65.85%

CAPACITY LOSS
−0.82%

TEMPORAL AT
65.03%

Removing spatial constraints (β_s=0) attenuates regression, as temporal dynamics are more transferable.

ADVANCED METHODS

Born-Again Networks & Knowledge Collapse

Iterative self-distillation MobileNet3D → MobileNet3D

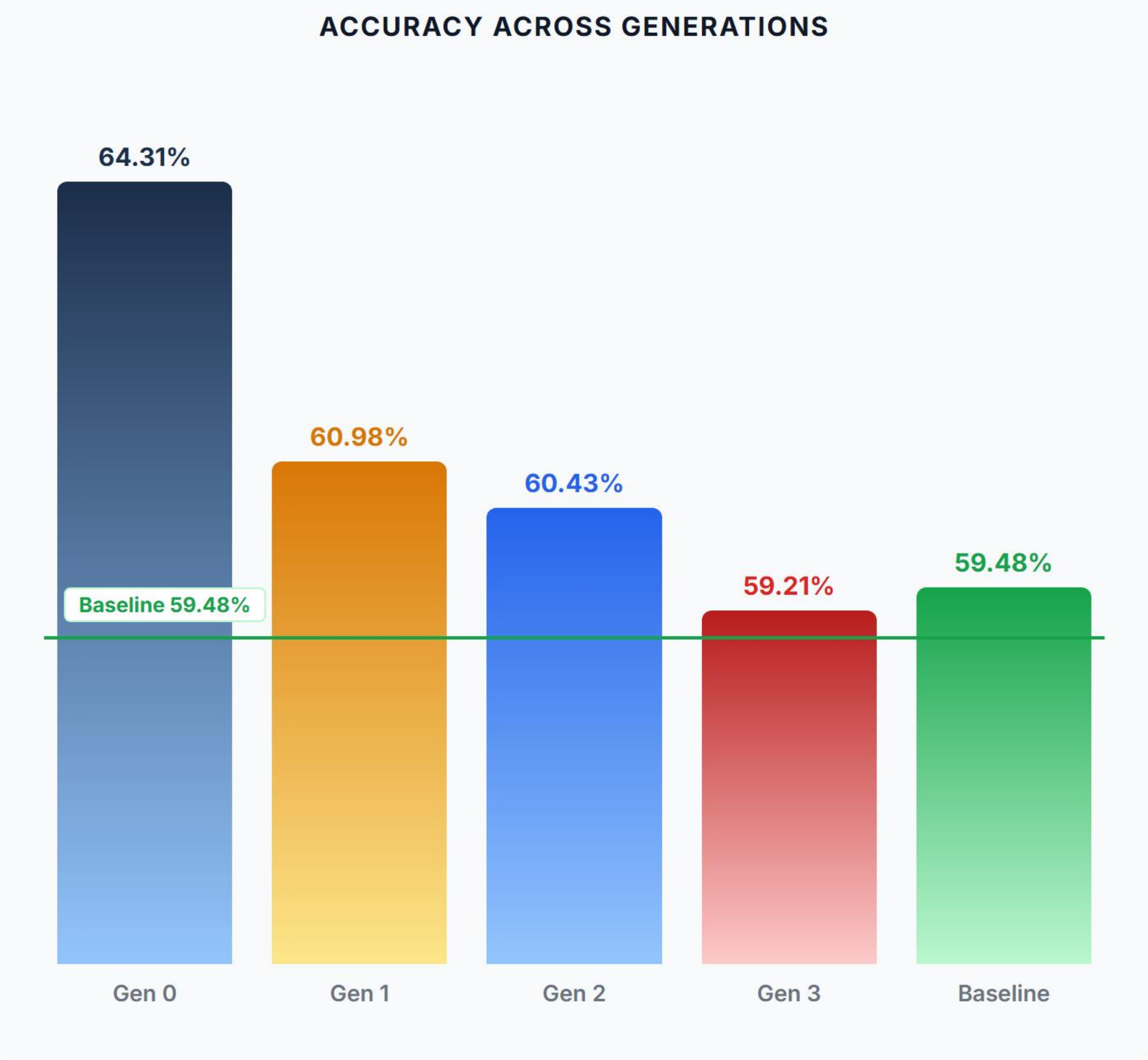
$\alpha = 0.5,\ T = 2.5$

Iterative Self-Distillation

GEN	TEACHER	T	A	ACCURACY
Gen 0	**3D ResNet-50**	**8.0**	**0.7**	**64.31%**
Gen 1	MobileNet3D Gen 0	2.5	0.5	**60.98%**
Gen 2	MobileNet3D Gen 1	2.5	0.5	**60.43%**
Gen 3	MobileNet3D Gen 2	2.5	0.5	**59.21%**
Baseline	No Teacher (Scratch)	—	—	59.48%

Considerations

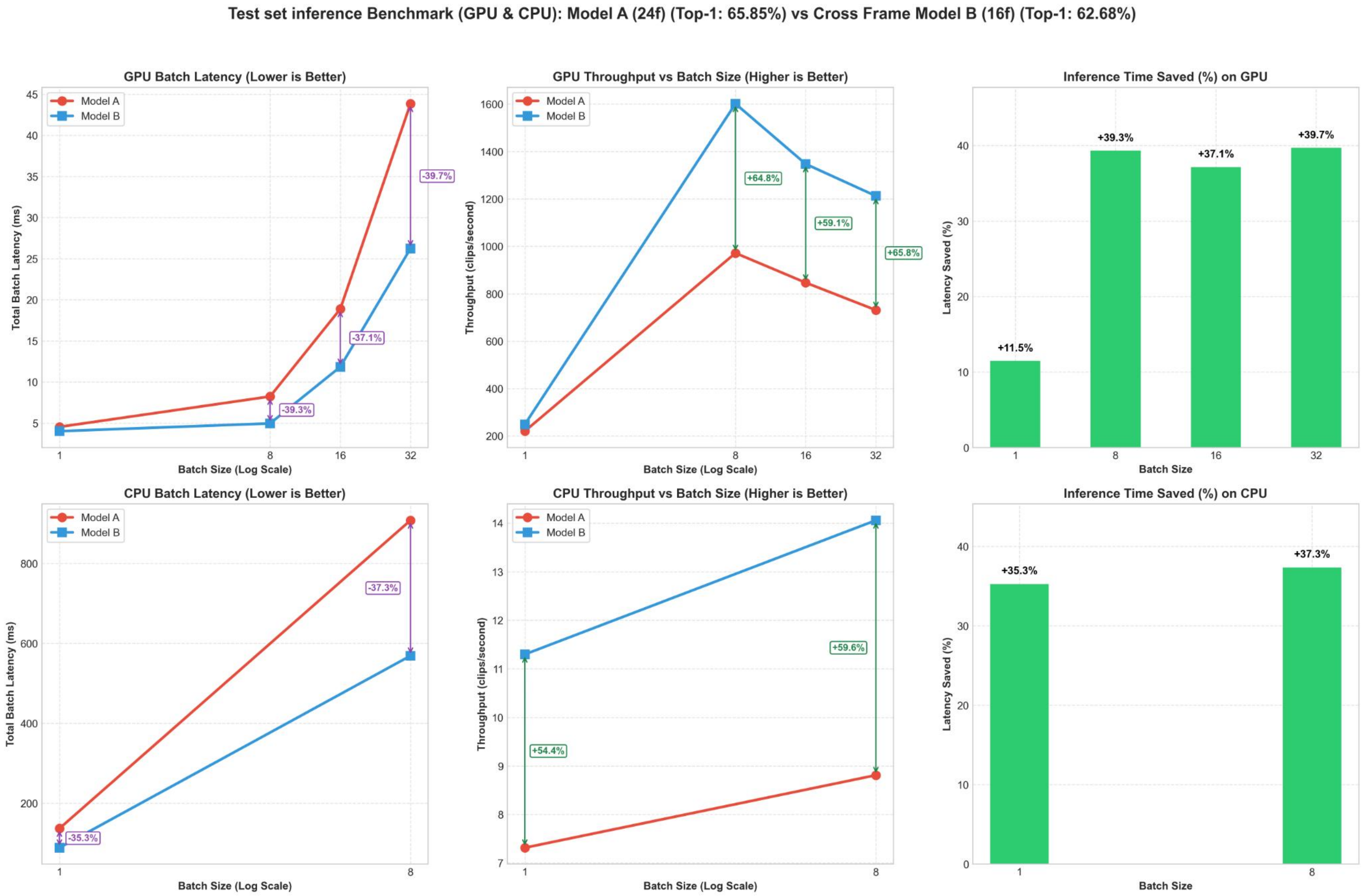
A compressed student (64.31%) lacks the visual representational capacity to act as an effective teacher. Distilling at **T=2.5** fails to prevent error propagation, causing progressive degradation back to baseline (59.48%).



EFFICIENCY & EDGE DEPLOYMENT

Cross-Frame Distillation

Asymmetric temporal sub-sampling (T=20, a=0.7) — 16f Student vs 24f Teacher



PERFORMANCE GAP

−3.17%

65.85% (24f) → 62.68% (16f)

Minimal accuracy cost for temporal compression

INFERENCE SPEEDUP

2.11×

CPU latency halved (-52.6%)

GPU throughput boosted by +68.8%

CONCLUSIONS

Key Takeaways & Future Work

NO FRAME DIST. (24F)	CROSS-FRAME DIST. (16F)
ACCURACY **65.85%** +6.37 pp vs. 59.48% Baseline	ACCURACY **62.68%** +3.20 pp vs. 59.48% Baseline
CPU LATENCY **53.52 ms** 24-frame computational load	CPU LATENCY **25.36 ms** -52.6% (2.11×) Latency halved on Edge CPU
GPU THROUGHPUT **991 v/s** Standard throughput	GPU THROUGHPUT **1673 v/s** +68.8% Register & Tensor Core alignment

Lessons Learned

The spatial capacity gap limits intermediate feature transfer. Logit distillation at high temperature (T=20) is the most stable guide.

Key Limitations

- (KD) High temperature (T=20) over-smooths fast, cyclic actions.
- (AT) Spatial capacity gap limits intermediate Feature Distillation.
- (BAN) Error propagation & bias accumulation in Born-Again Networks.

Future Work

- **Multi-Scale Temporal Sampling:** Alternate global segmented sampling with local, tight-stride windows to capture both long-term history and fine-grained cyclic actions.
- **Flexible Feature Alignment:** Introduce temporary learnable adapter blocks (e.g. 1×1×1 convs) during training to bridge the AT capacity gap.

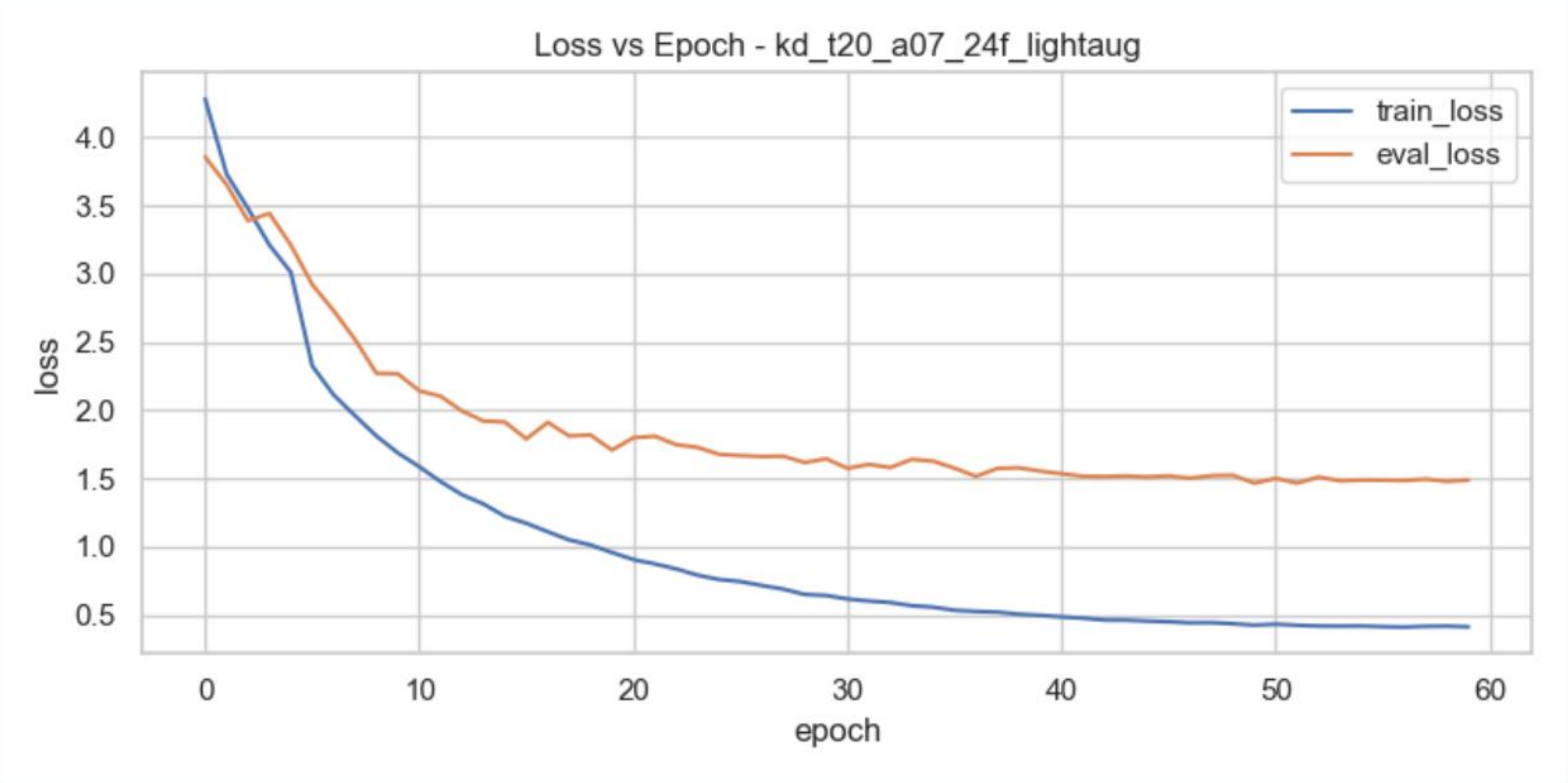
ADDITIONAL MATERIAL

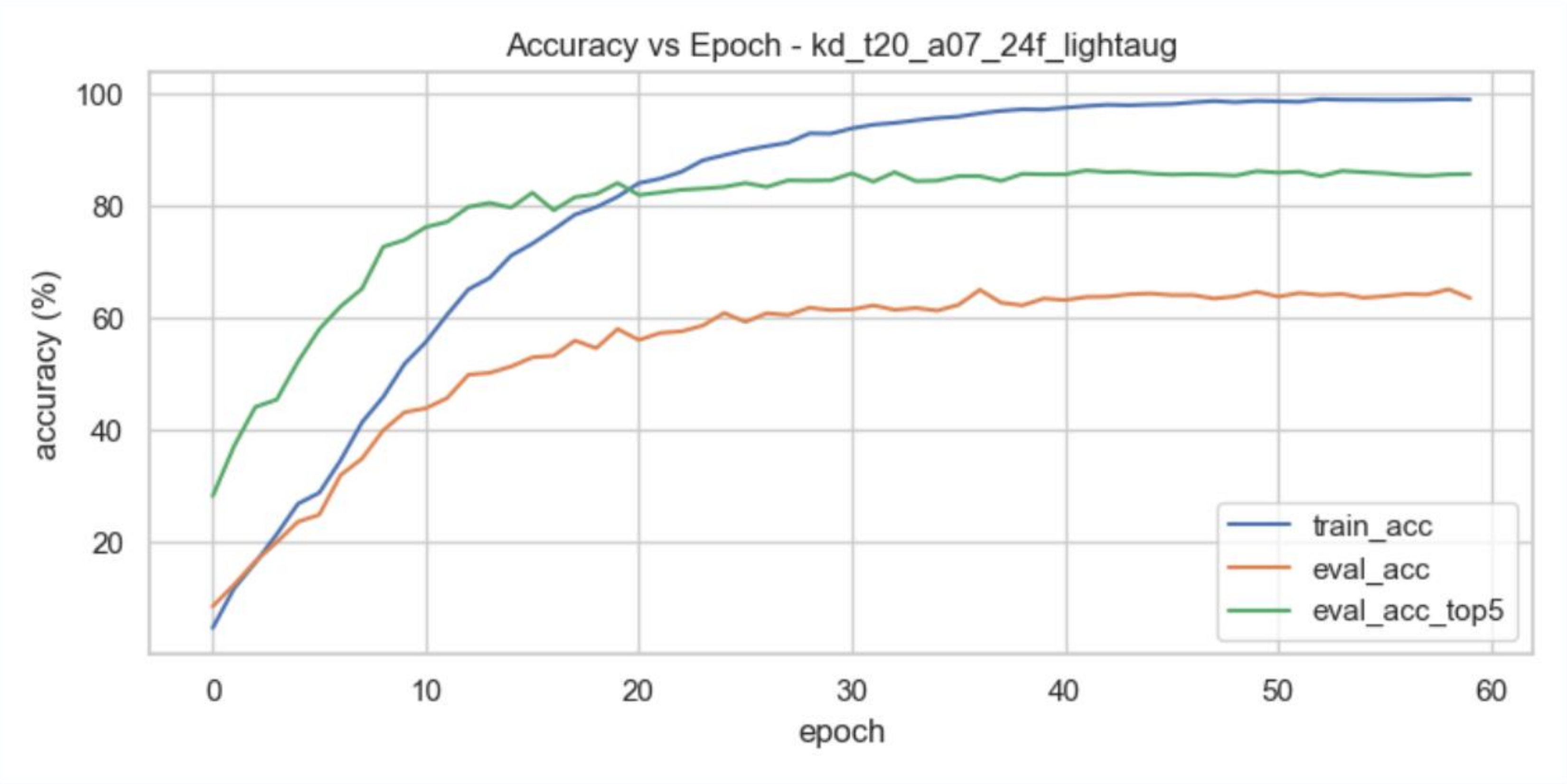
Backup Slides

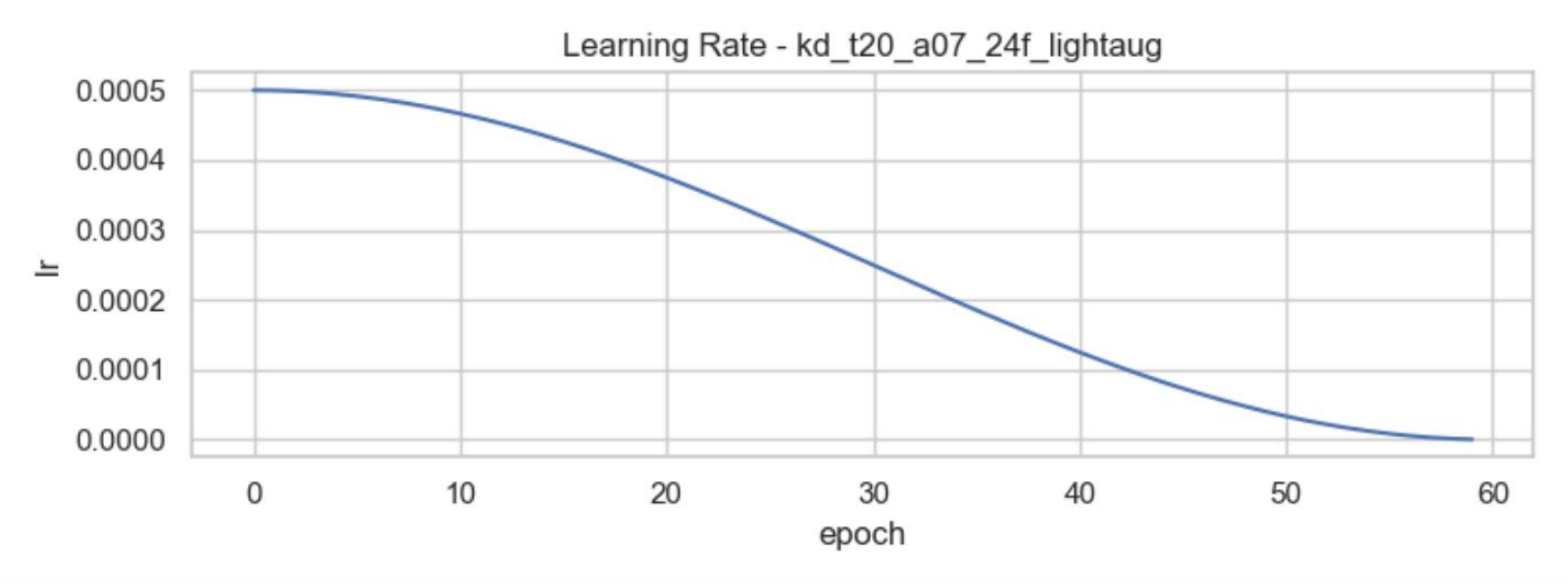
Quantitative plots, qualitative detailed analyses, and CM predictions.

Training Curves — KD T=20, a=0.7

Loss, accuracy, and LR schedule over 60 epochs







Stability & Convergence

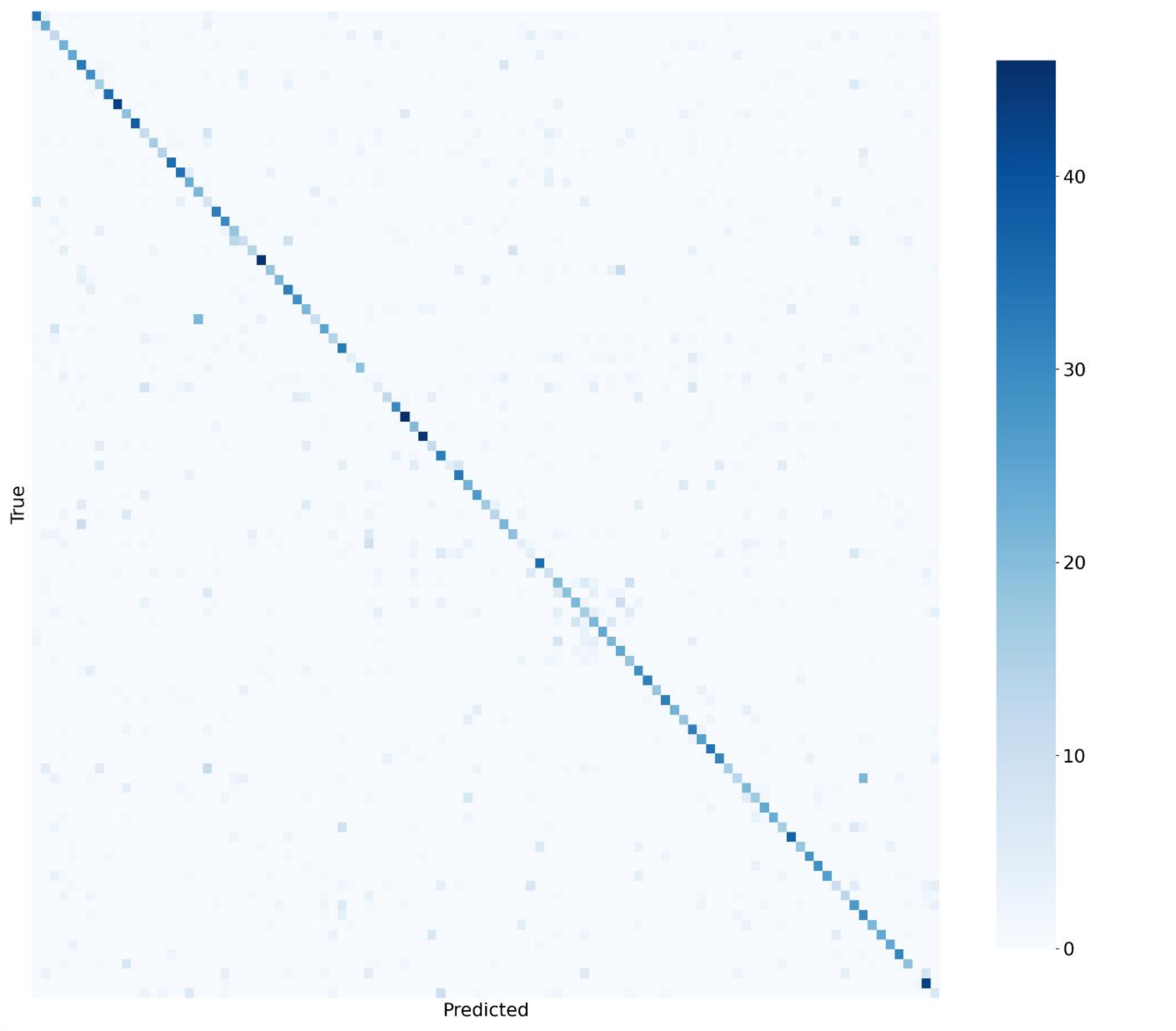
Stable training over 60 epochs. First 10 epochs serve as warmup with hard targets before soft distillation loss activates.

BACKUP - ERROR ANALYSIS

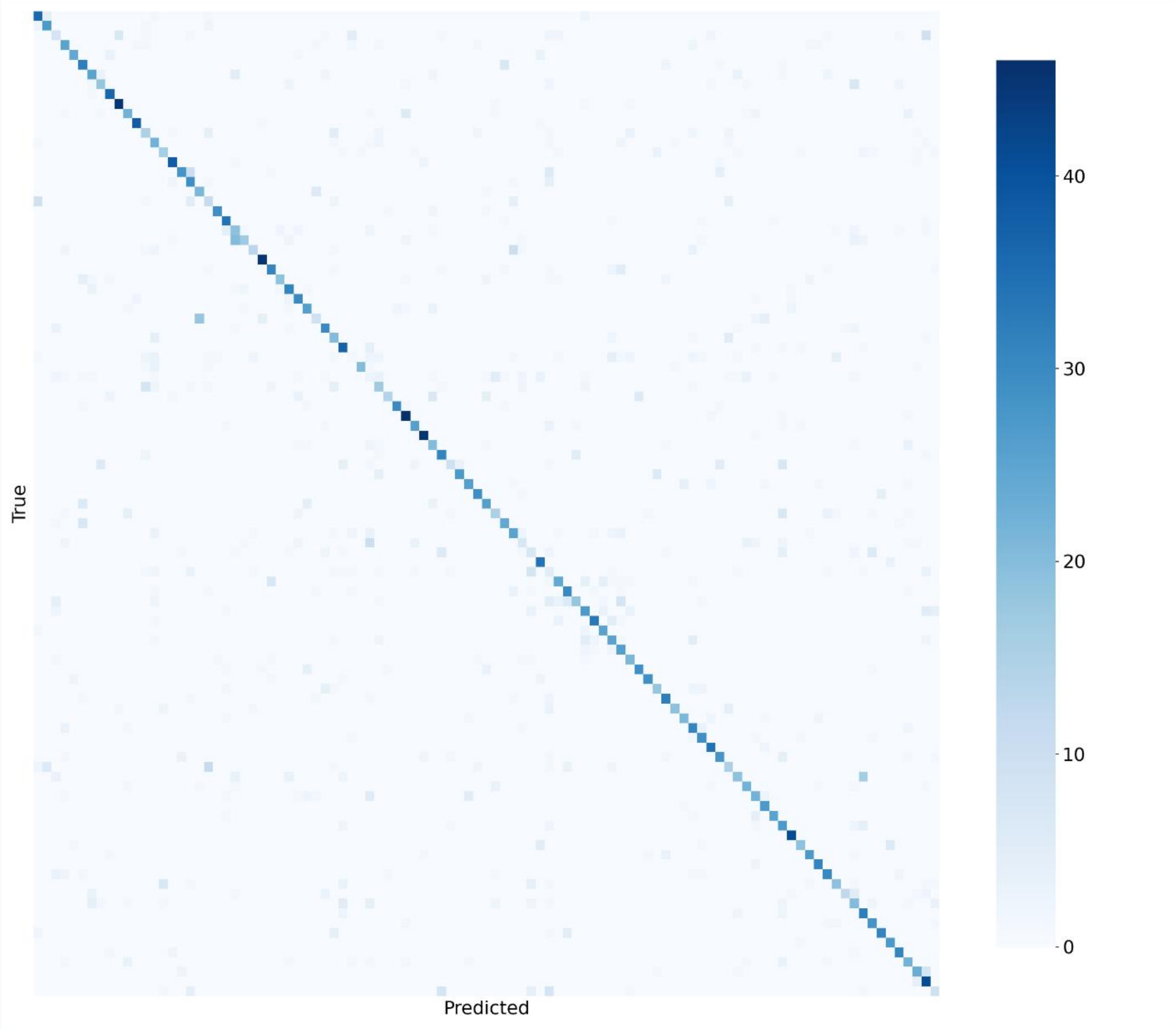
Confusion Matrix & Error Analysis

Baseline vs KD T=20 - Full normalized matrices

BASELINE - 59.48%



KD T=20 - 65.85%



Attention Transfer Loss: Before and After

Decomposing 3D video features to resolve the spatial-temporal collapse

Before: Single Collapsed Loss

$$\mathcal{L}_{AT} = \sum_l || normalize(mean_c(F^2))_{flat} - ... ||^2$$

- **Block Mismatch:** Hooked ResNet stages **[3, 4, 5]**. Stage 5 is the classifier head (produces 2D logits `[B, 101]`), which caused runtime crashes and zero gradient contribution.
- **1D Interpolation Mismatch:** Resizing a flattened 1D vector of shape `[T × H × W]` treats space and time as a single coordinate line. This mathematically mixed pixels across frame boundaries and unrelated spatial points, destroying feature geometry.
- **Temporal Signal Dilution:** Normalizing the entire flattened space meant temporal peaks were washed out by spatial dimensions.

59.14%

After: Spatial & Temporal Decomposition

$$\mathcal{L}_{AT} = \beta_s \mathcal{L}_{spatial} + \beta_t \mathcal{L}_{temporal}$$

- **Correct Block Pairing:** Hooked ResNet stages **[2, 3, 4]** (intermediate 3D features: `[B, C, T, H, W]`), correctly matching student stages **[2, 4, 6]**.
- **2D Bilinear Interpolation:** Spatial attention is calculated as `$mean_{c,t}(F^2)$` and resized spatially via 2D bilinear interpolation, preserving frame layout.
- **Temporal AT Isolation:** Temporal attention is calculated as `$mean_{c,h,w}(F^2)$` yielding `[B, T]`, which is transferred directly without any interpolation noise.

64.58% · 65.03%

Symmetric: β_s=0.05, β_t=0.05 · Temporal: β_s=0, β_t=0.10

Analysis of Attention Transfer Limitations

Theoretical barriers of classical attention transfer in lightweight student training

1. Attention Loss & Information Collapse

Key Bottleneck: Channel Pooling

To align mismatched feature map dimensions, classical AT averages activation maps along the channel axis. This mathematical flattening results in a severe loss of representation:

- **Feature Flattening:** Collapsing the channel dimension merges specialized filters (textures, edges, motion directions) into a generic spatial-temporal map.
- **Loss of Semantics:** The student is forced to learn *where* or *when* to look, but misses the underlying semantic explanation.
- **Coarse Regularization:** The student mimics a highly-abstracted projection rather than the rich visual filters that drive the teacher's decisions.

2. Capacity Gap & Structural Mismatch

Key Conflict: Geometric Incompatibility

The student model's architectural constraints limit its capacity to mimic the teacher's representations due to fundamental differences in conv designs:

- **Convolutional Disparity:** The teacher utilizes standard 3D convolutions that extract joint space-time-channel correlations.
- **Depthwise Factorization:** The student (MobileNet3D) uses Depthwise Separable Convolutions, which strictly isolate spatial-temporal filters from channel mixtures.
- **Gradient Conflict:** Forcing the student to replicate complex attention maps creates an over-constrained optimization. This leads to conflicting gradients (noise) that degrade overall convergence.